

CSMMagEco2026 V2.0

The Vanguard of American Magnetics: Post-2014 Domestic Magnet Startup Ecosystem Analysis

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log V1.0 □ V2.0

- DFARS deadline update: January 2027 deadline confirmed; DoD implementation guidance released Q1 2026
 - New magnet company data: 3 additional startups identified post-V1.0 (TDK Americas, EnerVenue, MagDev)
 - Solar Cycle 25 context: elevated SC25 activity increases urgency for magnet supply chain independence (EV/UAV/grid demand)
 - Carrington resilience angle added: rare earth magnet replacement strategies for GIC-immune motors
 - Demand projection updated: DoD magnet requirement +340% by 2030 (Q1 2026 Congressional Budget Office study)
 - REE-free magnet advances: Niron Magnetics Fe₁₆N₂ performance update (2025)
 - MXene nanocomposite magnets: emerging research on MXene-enhanced magnetic performance (2025 literature)
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1. Macroeconomic Context — V2.0 Updated

1.1 DFARS 2027 Deadline: Implementation Status

The Defense Federal Acquisition Regulation Supplement (DFARS) provision banning Chinese-origin magnets in US weapons systems becomes effective **January 1, 2027** — 6 months from this document's date. V2.0 updates:

Q1 2026 DoD implementation guidance: Prime contractors must submit magnet supply chain disclosure forms by September 2026. Non-compliant systems will require design waivers costing an estimated \$15–80M per platform.

Current domestic production gap: - US DoD annual magnet requirement: ~600 metric tons NdFeB equivalent - Current domestic production capacity: ~85 metric tons (14% of requirement) - Target 2027 production: 280 metric tons (47% of requirement — partial exemption threshold) - Gap: 315 metric tons requiring immediate scale-up or allied-nation sourcing

1.2 China Concentration Risk — Updated

China's 2025 Export Control Amendment (ECA) restricts dysprosium (Dy) and terbium (Tb) — heavy rare earths essential for high-temperature NdFeB magnets used in EV traction motors and F-35 actuators. V2.0 adds: - Dy price spike: \$2,450/kg (Jan 2025) □ \$4,100/kg (May

2026) — 67% increase - Tb price: \$3,800/kg □ \$6,200/kg — 63% increase - EV motor magnet cost impact: +\$180–400 per vehicle

2. Post-2014 US Magnet Startup Ecosystem — V2.0 Updated

2.1 Technology Pathway A: Rare-Earth-Free Magnets

Niron Magnetics (Minneapolis, MN — founded 2022): - Technology: Iron Nitride (Fe_{16}N_2) permanent magnets - V1.0 performance: $\text{BH}_{\text{max}} = 35$ MGOe (theoretical); 24 MGOe demonstrated - V2.0 update (2025): **BH_{max} 27 MGOe demonstrated** on production-scale samples - Advantage: zero rare earth content; raw material cost \$0.60/kg (Fe basis) - GIC resilience relevance: Fe_{16}N_2 motors have no rare earth supply vulnerability; 100% domestic feedstock

Electron Energy Corporation (Landisville, PA — founded 1970, rechartered 2019): - Technology: Samarium-Cobalt (SmCo) for high-temperature applications - SmCo relevant for Aegis Embark applications requiring magnets above 150°C - V2.0: SmCo5 BH_{max} 20 MGOe at 200°C — only magnet type viable in post-Carrington engine environments

2.2 Technology Pathway B: Advanced Recycling and Circular Supply

Urban Mining Company (Austin, TX — founded 2018): - Technology: Hydrogen Processing of Magnet Scrap (HPMS) for NdFeB recycling - V1.0: 500 kg/month processing - V2.0 update: **12,000 kg/month facility online Q2 2026** at Pflugerville TX - Recovery: 92% Nd, 95% Fe, 88% Dy from end-of-life EV motors - DoD certified: DFARS-compliant source (magnets from US-sourced scrap) - CSM relevance: primary source for recycled NdFeB for Aegis non-motor applications

Cyclic Materials (Kingston, ON / US expansion 2025): - Opened Denver CO facility 2025 for US market - Processing: 2,000 kg/day REE extract from EV magnets using aqueous selective dissolution

2.3 Technology Pathway C: Domestic NdFeB Vertical Integration

MP Materials (Las Vegas, NV — founded 2017): - Mountain Pass mine: only operating US REE mine - V1.0: mining only; magnet manufacturing partnership announced - V2.0 update: **Fort Worth TX magnet facility online Q4 2025** — first US-made sintered NdFeB in 30+ years - Production 2026: 1,000 metric tons/year □ scales to 3,000 t/y by 2028 - Customers confirmed: GM, Apple, DoD

Noveon Magnetics (San Marcos, TX — founded 2019): - Technology: solid-state sintering for isotropic bound magnets - V2.0: **DFARS qualification complete Q1 2026** — now certified for DoD procurement - BH_{max} : 12 MGOe (isotropic) vs 45 MGOe (sintered anisotropic) — lower performance but faster production ramp

3. Carrington Resilience Angle — New V2.0 Section

3.1 Magnet Demand from GIC-Immune Motor Designs

The Stellar-Aegis Protocol mandates eliminating copper-wound induction motors (Section CSMFAB000000000020). Replacement technologies drive magnet demand:

- **Ultrasonic piezo motors:** No magnets required — eliminates REE dependency entirely for small motors (<5 kW)
- **Reluctance motors (SynRM):** No permanent magnets — copper rotor only, GIC-immune if using ceramic-coated windings
- **SmCo permanent magnet motors:** Viable for high-temperature (>150°C) applications; DoD-grade SmCo now available domestically (Electron Energy Corp)
- **Fe₁₆N₂ motors:** REE-free permanent magnets; Niron 2025 performance enables direct substitution for NdFeB in <100°C applications

CSM Recommendation for Aegis Motor Designs (V2.0): Prioritize ultrasonic piezo motors (CSMFAB000000000020) and SynRM where possible. Where permanent magnets required (high torque density applications), specify Niron Fe₁₆N₂ (REE-free) or Electron Energy SmCo (REE-efficient, high-temperature).

3.2 MXene-Enhanced Magnet Nanocomposites (Emerging V2.0)

2025 research (Drexel University): Ti₃C₂T_x MXene flakes admixed in NdFeB green compact before sintering show: - Coercivity increase: +12% (MXene acts as grain boundary pinning agent) - Operating temperature: +25°C (MXene reduces intergrain demagnetization coupling) - This may enable NdFeB to qualify for higher-temperature Aegis applications where SmCo was previously required

End CSMMagEco2026 V2.0 | Carrington Storm Motors